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Chapter 4 Numerical characteristics of random variables

Section 2 Variance

- Definition of variance
- Variance of discrete random variables
- Variance of continuous random variables
- Properties of the variance

Definition of variance

The variance measures the spread of the distribution around the mean. An r.v. with a zero variance is constant. The larger the variance, the more uncertain the r.v. is, in the mean square sense.

Definition of the variance If X is an r.v. with $E[X] = \mu_X$, then $Var(X)$ or $D(X)$, called the variance of X, is defined by

$$Var(X) = D(X) = E\left[(X - \mu_X)^2\right]$$

and the Standard Deviation (SD) σ_X of X is

$$\sigma_X = \sqrt{Var(X)} = \sqrt{D(X)}$$

The square is used to ensure that all differences X-E(X) have a positive effect

Also, one may use σ^2 or $\sigma_X{}^2$ for the variance of the r.v. X. The variance of X is defined as the average square distance between X and its mean μX . The standard deviation is used more typically since the units associated with SD are the same as the r.v. X itself, whereas the variance is measured in squared units of X .

Definition of variance

$Var(\mathrm{X})$ is a real number.

- Formally, it is the sum of the product of the squared deviation of each value of X and their average value and the corresponding probability.
- In essence, it reflects the degree of deviation of X around the "average value", so it is a sign to measure the degree of dispersion of the value of X.
- Physically represents the moment of inertia of a particle system passing through the longitudinal axis of the center of gravity E(X).

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Variance of discrete random variables

1. **Definition.** Suppose the distribution law of the discrete random variable X is:

X	x_0	x_1	$x_2 \cdots\cdots$	$x_n \cdots\cdots$
P_K	p_0	p_1	$p_2 \cdots\cdots$	$p_n \cdots\cdots$

If the series $\sum_{k=1}^{\infty}[x_k - E(X)]^2 p_k$ absolutely converges, then this series is called the variance of X, which is recorded as:

$$D(X) = Var(X) = \sum_{k=1}^{\infty}[x_k - E(X)]^2 p_k = E[X - E(X)]^2$$

Note. If the series $\sum_{k=1}^{\infty}[x_k - E(X)]^2 p_k$ does not absolutely converge, it is said that Var(X) does not exist.

Variance of discrete random variables

2. The variance of several common distributions

（1） （0-1） distribution

If the random variable X can only take two values, 0 and 1, its distribution law is:

$$P(X=k)=p^k(1-p)^{(1-k)} \qquad k=0,1. \quad 0<p<1$$

Then: $Var(\mathrm{X})=(0-p)^2\cdot q+(1-p)^2\cdot p=pq$ $\mathrm{E(X)}=p$

（2） Binomial distribution

Assuming that the random variable X obeys the binomial distribution with parameters (n, p), that is X~B(n,p), its distribution law is:

$$P(X=k)=c_n^k p^k q^{n-k}, \quad k=1,2,\cdots,n$$

Then: $Var(\mathrm{X})=npq$ $\mathrm{E(X)}=np$

Variance of discrete random variables

（3） Poisson distribution

If all the possible values of the random variable X are: 0, 1, 2... and its distribution law (each probability of its value) is:

$$P(X=k)=\frac{\lambda^{k}e^{-\lambda}}{k!} \quad k=0,1,2,\cdots \qquad X\sim P(\lambda)$$

Then: $Var(\mathrm{X})=\lambda$ $\mathrm{E}(\mathrm{X})=\lambda$

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Variance of continuous random variables

1. Definition.

Suppose the probability density of continuous random variable X is f(x).

If the integral $\int_{-\infty}^{+\infty} [x - E(X)]^2 f(x)dx$ is absolutely convergent, then this integral is called the variance of X, which is recorded as:

$$D(X) = Var(X) = \int_{-\infty}^{+\infty} [x - E(X)]^2 f(x)dx$$

$$= E[X - E(X)]^2$$

Variance of continuous random variables

2. The variance of several common distributions

(1) Uniform Distribution

If the continuous random variable X has a probability density f (x):

$$f(x) = \begin{cases} \dfrac{1}{b-a} & a < x < b \\ 0 & else \end{cases} \qquad X \sim U\,[a, b]$$

Then: $Var(X) = \dfrac{(b-a)^2}{12}$ $\qquad E(X) = \dfrac{a+b}{2}$

Variance of continuous random variables

(2) Exponential Distribution

If the continuous random variable X has a probability density f (x):

$$f(x) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}} & x > 0 \\ 0 & other \end{cases}$$

Where $\theta > 0$ is a constant

Then: $Var(X) = \int_{-\infty}^{+\infty} [x - E(X)]^2 f(x) dx$

$$= \int_{0}^{+\infty} (x-\theta)^2 \cdot \frac{1}{\theta} e^{-\frac{x}{\theta}} dx \quad = \theta^2$$

Which is $Var(X) = \theta^2$ $E(X) = \theta$

Variance of continuous random variables

(3) Normal Distribution

If the probability density f (x) of the random variable X is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty \qquad X \sim N(\mu, \sigma^2)$$

Then: $Var(X) = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x) dx$

$$= \int_{-\infty}^{+\infty} (x-\mu)^2 \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

令: $z = \frac{x-\mu}{\sigma} \quad \Rightarrow$

Variance of continuous random variables

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \sigma^2 z^2 \frac{1}{\sigma} e^{-\frac{z^2}{2}} \sigma \, dz \qquad = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} z^2 e^{-\frac{z^2}{2}} dz$$

$$= \frac{\sigma^2}{\sqrt{2\pi}} \left[-z e^{-\frac{z^2}{2}} \Big|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} e^{-\frac{z^2}{2}} dz \right] = \frac{\sigma^2}{\sqrt{2\pi}} \cdot \sqrt{2\pi} \quad = \sigma^2$$

$Var(X) = \sigma^2$

$E(X) = \mu$

Conclusion: The parameter σ^2 of the density function in the normal distribution is exactly the variance of the random variable X.

The parameter μ of the density function in the normal distribution is exactly the mathematical expectation of the random variable X.

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The prosperities of variance

1. $Var(X) = E(X^2) - [E(X)]^2$

This is an VERY important property, and frequently used calculation formula

Prove: $\because \quad Var(X) = E[X - E(X)]^2$

$$= E[X^2 - 2XE(X) + (E(X))^2]$$

$$= E(X^2) - E[2XE(X)] + E[(E(X))^2]$$

$$= E(X^2) - 2E(X)E(X) + [E(X)]^2$$

$$= E(X^2) - [E(X)]^2$$

Note: This formula gives a formula for calculating the variance of a random variable X, and also gives the relationship between mathematical expectation and variance.

Properties of the variance

Example 1. Suppose X~U[a,b], that is, its probability density function is:

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 0, & else \end{cases}$$

Calculate the variance of X.

Answer: Because: $Var(X) = E(X^2) - [E(X)]^2$

$$E(X) = \frac{a+b}{2}$$

So: $$Var(X) = \int_a^b x^2 \cdot \frac{1}{b-a} dx - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{(b-a)^2}{12}$$

Properties of the variance

Example 2.

Assuming that the random variable X obeys a random geometric distribution, its distribution law is:

$$P(x = k) = p(1-p)^{k-1}, \quad k = 1,2,\cdots n \qquad 0 < p < 1$$

Calculate $Var(X)$

Answer: $q = (1-p)$

Infinitely shrinking geometric series summation formula

$$E(X) = \sum_{k=1}^{\infty} kpq^{k-1} = p \cdot \sum_{k=1}^{\infty} (q^k)'$$

$$= p \cdot (\sum_{k=1}^{\infty} q^k)' = p \cdot (\frac{q}{1-q})' = \frac{1}{p}$$

Exchange order of summation and derivation

Properties of the variance

$$E(X^2) = \sum_{k=1}^{\infty} k^2 p q^{k-1}$$

$$= p \cdot [\sum_{k=1}^{\infty} k(k-1)q^{k-1} + \sum_{k=1}^{\infty} k q^{k-1}]$$

$$= qp \cdot (\sum_{k=1}^{\infty} q^k)'' + E(X) = qp \cdot (\frac{q}{1-q})'' + \frac{1}{p}$$

$$= qp \cdot \frac{2}{(1-q)^3} + \frac{1}{p} = \frac{2q}{p^2} + \frac{1}{p} = \frac{2-p}{p^2}$$

So: $Var(X) = E(X^2) - [E(X)]^2$

$$= \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}$$

Properties of the variance

2. If c is a constant, then: $Var(c)=0$

The sufficient and necessary condition for $Var(X)=0$ is that X takes the constant c with probability 1, that is, **P(X=*c*)=1**,

Obviously, here $c = E(X)$.

3. If c is a constant and X is a random variable, then:

$$Var(cX) = c^2 Var(X)$$

Prove: $Var(cX) = E[(cX)^2] - [E(cX)]^2$

$$= E(c^2X^2) - [cE(X)]^2$$

$$= c^2E(X^2) - c^2[E(X)]^2$$

$$= c^2[E(X^2) - (E(X))^2] = c^2Var(X)$$

Properties of the variance

4. If X, Y are independent random variables, then:

$$Var(X+Y) = Var(X) + Var(Y)$$

Because X and Y are independent of each other, X-E(X) and Y-E(Y) are also independent of each other

Prove: $Var(X+Y) = E[(X+Y) - E(X+Y)]^2$

$$= E\{[X - E(X)] + [Y - E(Y)]\}^2$$

$$= E[X - E(X)]^2 + E[Y - E(Y)]^2 + \underline{2E\{[X - E(X)]}\underline{[(Y - E(Y)]\}}$$

$$= Var(X) + Var(Y) \qquad +2E[X - E(X)] \cdot E[Y - E(Y)]$$

$$= Var(X) + Var(Y) + 2[E(X) - E(X)] \cdot [E(Y) - E(Y)]$$

$$= Var(X) + Var(Y)$$

Note: This property can be extended to any finite sum of multiple independent random variables.

Properties of the variance

5. (Chebyshev‘s inequality) Assume that the random variable X has mathematical expectation $E(X) = \mu$, variance $Var(X) = \sigma^2$. For any positive number ε .

Inequality $\boldsymbol{P\{|X - \mu| \geq \varepsilon\} \leq \frac{\sigma^2}{\varepsilon^2}}$ holds.

Note:

- Another form of Chebysev's inequality:

$$\boldsymbol{P\{|X - \mu| < \varepsilon\} > 1 - \frac{\sigma^2}{\varepsilon^2}}$$

Properties of the variance

- The role of Chebyshev's inequality: Given the unknown distribution of random variable X, a method of estimating the lower limit of probability $\{|X-\mu| < \varepsilon\}$

  such as $\varepsilon = 3\sigma$, then:

$$P\{|X-E(X)| \geq 3\sigma\} \leq \frac{\sigma^2}{9\sigma^2} \approx 0.111$$

- It can be seen from Chebyshev's inequality that if σ^2 is smaller, the probability of event $\{|X-E(X)|<\varepsilon\}$ is greater, that is, the probability that the random variable X is concentrated near the expectation is greater.

- **The probability meaning of variance is derived from this:**

  it characterizes the degree of dispersion of random variables

Properties of the variance

Example 3.

It is known that the average number of white blood cells per milliliter in normal male adult blood is 7,300, and the mean square error is 700.

Try to use Chebyshev's inequality to estimate the probability that the number of white blood cells per milliliter is between 5200 and 9400.

Answer: Let X: the number of white blood cells per milliliter

$$E(X) = 7300, \quad Var(X) = 700^2$$

$$\begin{aligned} P(5200 \le X \le 9400) &= P(5200 - 7300 \le X - 7300 \le 9400 - 7300) \\ &= P(-2100 \le X - E(X) \le 2100) \\ &= P(|X - E(X)| \le 2100) \end{aligned}$$

From Chebyshev's inequality: $\Rightarrow$

Properties of the variance

$$P\{|X - E(X)| \leq 2100\} \geq 1 - \frac{Var(X)}{(2100)^2}$$

$$= 1 - \left(\frac{700}{2100}\right)^2 \quad = 1 - \frac{1}{9} = \frac{8}{9}$$

That's means, the probability that the number of white blood cells per milliliter is between 5200 and 9400 is not less than 8/9.

Example 4. In each experiment, the probability of event A occurring is 0.75. Use Chebyshev's inequality to find: how big *n* needs to be so that in *n* independent repeated trials, the probability of event A occurring between 0.74 and 0.76 is at least 0.90?

Answer: Suppose X: the number of occurrences of event A in n trials.

Properties of the variance

$$X \sim B\,(n,\ 0.75)$$

$$E(X) = 0.75\ n, Var(X) = 0.75\ \times 0.25n = 0.1875n$$

It is necessary to find the minimum n that satisfying to condition:

$$P(0.74 < \frac{X}{n} < 0.76) \geq 0.90$$

$$P(0.74 < \frac{X}{n} < 0.76)\ = P(0.74n < X < 0.76n)$$

$$= P(0.74n - 0.75\ n < X - 0.75n < 0.76n - 0.75\ n)$$

$$= P(-0.01\ n < X - 0.75n < 0.01n)$$

$$= P(|X - E(X)| \leq 0.01n)$$

The probability that the frequency of occurrence of event A is between 0.74 and 0.76 is at least 0.90

Take in Chebyshev's inequality $\varepsilon = 0.01\ n$:

Properties of the variance

Then there are: $P(0.74 < \frac{X}{n} < 0.76) = P(|X - E(X)| \leq 0.01n)$

$$\geq 1 - \frac{Var(X)}{(0.01n)^2} = 1 - \frac{0.1875n}{0.0001n^2} = 1 - \frac{1875}{n}$$

Let: $1 - \frac{1875}{n} \geq 0.9$

$$n \geq \frac{1875}{1 - 0.9} = 18750$$

Properties of the variance

Example 6. Assume the mathematical expectation of X is E(X), and the variance is Var(X). **Calculate:** The mathematical expectation and variance of random variable

$$Y = \frac{X - E(X)}{\sqrt{Var(X)}}$$

.

Answer:

$$E(Y) = E\left(\frac{X - E(X)}{\sqrt{Var(X)}}\right) = \frac{1}{\sqrt{Var(X)}} E[X - E(X)]$$

$$= \frac{1}{\sqrt{Var(X)}} [E(X) - E(X)] = 0$$

$$Var(Y) = Var\left(\frac{X - E(X)}{\sqrt{Var(X)}}\right) = \frac{1}{Var(X)} Var[X - E(X)]$$

$$= \frac{Var(X)}{Var(X)} = 1$$

Y is a standardized random variable